Genzero

POST COMPLETION

LCA RESULTS

May 2025

WAUGH THISTLETON

ARCHITECTS

1.1 PRACTICAL CLASSROOM

LOCATION : Hartlepool SEN

GIA : 129 m²

Delivered at the SEND school in Hartlepool, the standard classroom suite will enable the school to enhance its offer to vulnerable children by locating the accommodation adjacent to the horticultural and animal space. This supports outdoor learning for children of all ages. This alternative space will provide a base for children to develop the wider caring side of their curriculum. The pod is designed to be self-sufficient, with the ability to accommodate wet practical activities and direct access to the outdoor environment. The suite is constructed using both volumetric modular and panellised construction, allowing both methods and their interaction to be developed and tested. Using UK-grown timber, the suite demonstrates the benefits of fast construction using high-quality materials that are durable and require little maintenance.

OPERATIONAL CARBON

Operational carbon refers to the carbon emissions produced by a building or infrastructure during its use and operation. This includes emissions from energy used for heating, cooling, ventilation, lighting, and other operational needs.

Energy Use Intensity
41.0 kWh/yr/m²

Number PVs
27 (not installed)

PVs Annual Generation
87.1 kWh/yr/m² (not installed)

Natural ventilation
8 l/s/p

EMBODIED CARBON

Embodied or embedded carbon is the carbon dioxide emitted in producing materials and products. It is a total of the energy used to extract and transport raw materials in addition to emissions associated with the manufacturing process. In construction embodied carbon is measured over the whole lifespan of a building, typically assumed to be 60 years. The carbon impact of the building's construction is calculated at each phase of the development chronologically. A1-A5 Upfront, production and construction, B1-B6 in use, and C1-C4 End of life. The carbon emissions associated with each stage are reported as kgCO2e/m². Biogenic carbon is the carbon stored within the timber elements. recycling or disposal as waste. The carbon emissions associated with each stage are reported as kgCO2e/m².

Substructure
90

Structure
99

Envelope
37

Internal fit out
33

MEP
77

A1-A5 Carbon

A1-A5

336
kgC02e/m²

UPFRONT EMBODIED CARBON

A1-A5 of 336 kgCO2e/m² excluding sequestration is less than the required DfE target of 525kgCo2e/m2 and 325 kgCo2e/m2 including sequestration.

A-C

568
kgC02e/m²

WHOLE LIFE EMBODIED CARBON

A-C including sequestered of 568 kgCO2e/m².
This is less than the 2025 RIBA A-C target for completed buildings of <675 kgCO2e/m².

-114
kgC02e/m²

BIOGENIC SEQUESTERED CARBON

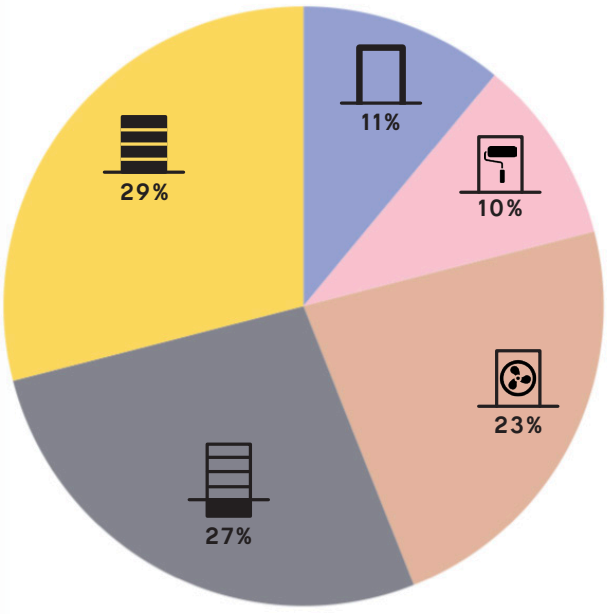
There is 15 TCO2e biogenic carbon sequestered within the fabric of Practical Classroom, or 114 kgCO2e/m². This is equivalent to 34% of the upfront embodied carbon.

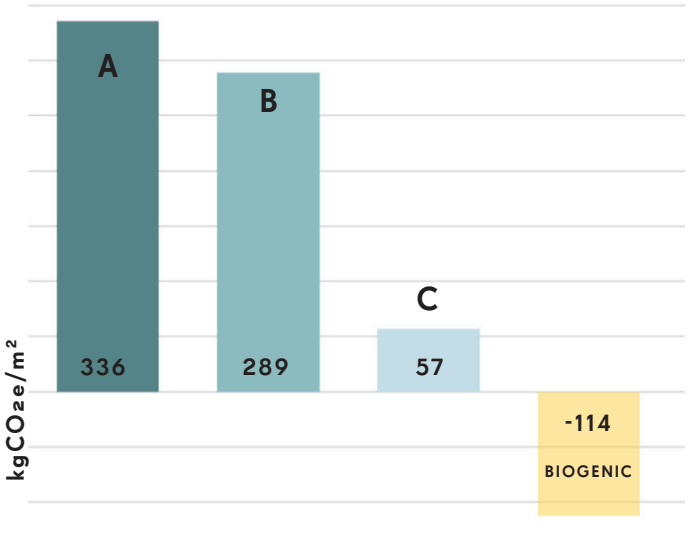
DEMAND

41
kWh/m²/y

OPERATIONAL ENERGY

The annual operational energy demand for the Practical Classroom is 41.0 kWh/m²/y.
This is less than the 2030 RIBA target for educational buildings of <60 kWh/m²/y and less than the DfE Technical Annex J Target of 61 kwh/yr/m²





TOTAL = 568 kgco2e/m²

Figure: Breakdown of upfront carbon A1-A5 per building elements & Breakdown of whole life carbon A-C

1.2 STEM

LOCATION : Belvedere School

GIA : 276 m²

The two-storey STEM building is constructed adjacent to the existing sports block and playing fields at Belvedere School. It was determined that providing a general or sports classroom on the ground floor and a STEM facility on the first floor would enhance the sports offer by integrating theory and science elements into the Physical Education curriculum.

The direct access links to the playing field and the adjacency to the existing sports hall—which does not have any similar spaces—complement the accommodation of this block. The Specialist Suite is delivered using prefabricated timber cassettes made from UK-grown timber and demonstrates that the DfEs carbon targets are achievable. The building tests the viability and suitability of natural ventilation which also demonstrates a reduced operational energy requirement. The building has the potential to be net zero in operation if supplied with PVs

OPERATIONAL CARBON

Operational carbon refers to the carbon emissions produced by a building or infrastructure during its use and operation. This includes emissions from energy used for heating, cooling, ventilation, lighting, and other operational needs.

Energy Use Intensity
37.4 kWh/yr/m²

Number PVs
36 (not installed)

PVs Annual Generation
43.5 kWh/yr/m² (not installed)

Natural ventilation
8 l/s/p

EMBODIED CARBON

Embodied or embedded carbon is the carbon dioxide emitted in producing materials and products. It is a total of the energy used to extract and transport raw materials in addition to emissions associated with the manufacturing process. In construction embodied carbon is measured over the whole lifespan of a building, typically assumed to be 60 years. The carbon impact of the building's construction is calculated at each phase of the development chronologically. A1-A5 Upfront, production and construction, B1-B6 in use, and C1-C4 End of life. The carbon emissions associated with each stage are reported as kgCO2e/m². Biogenic carbon is the carbon stored within the timber elements. recycling or disposal as waste. The carbon emissions associated with each stage are reported as kgCO2e/m².

Substructure
102

Structure
143

Envelope
79

Internal fit out
40

MEP
91

A1-A5 Carbon

A1-A5

455

kgC02e/m²

UPFRONT EMBODIED CARBON

A1-A5 of 455 kgCO2e/m² excluding sequestration and 252 kgCO2e/m² including sequestration is less than the required DfE target of 525kgCo2e/m2 and 325 kgCo2e/m2 including sequestration

A-C

689

kgC02e/m²

WHOLE LIFE EMBODIED CARBON

A-C including sequestered of 689 kgCO2e/m².

This is slightly higher than the 2025 RIBA A-C target for completed buildings of <675 kgCO2e/m².

-249

kgC02e/m²

BIOGENIC SEQUESTERED CARBON

There is 70 TCO2e biogenic carbon sequestered within the fabric of the STEM building, or 249 kgCO2e/m²This is equivalent to 54% of the upfront embodied carbon.

DEMAND

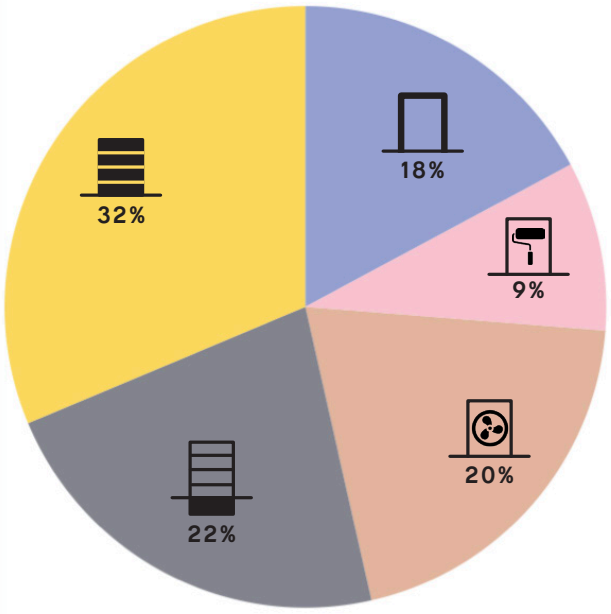
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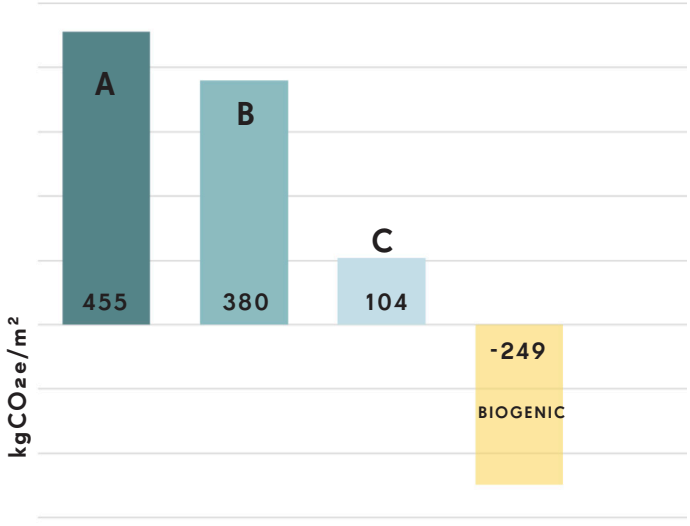
kWh/m²/y

OPERATIONAL ENERGY

The annual operational energy demand for the STEM building is 37.4 kWh/m²/y.

This is less than the 2030 RIBA target for educational buildings of <60 kWh/m²/y and less than the DfE Technical Annex J Target of 61 kwh/yr/m²





TOTAL = 689 kgco2e/m²

Figure: Breakdown of upfront carbon A1-A5 per building elements & Breakdown of whole life carbon A-C

1.3 SPORTS CHANGING FACILITY

LOCATION : West Coventry Academy GIA : 342 m²

Constructed at West Coventry Academy the Sports Changing Suite supports their existing levels of outdoor sports facilities. By locating this alternative changing accommodations at the heard of the outdoor sports areas it will be able to increase the offer of grass field based sports and the school can offer pupils the opportunity of created a community sports hub with the additional staff room and covered outdoor teaching space.

The suite is delivered using volumetric construction, incorporating a hybrid timber and steel frame. An Energy Pod is also included as part of the project, demonstrating how school IT and plumbing services can be upgraded quickly and efficiently using prefabricated volumetric solutions. T

OPERATIONAL CARBON

Operational carbon refers to the carbon emissions produced by a building or infrastructure during its use and operation. This includes emissions from energy used for heating, cooling, ventilation, lighting, and other operational needs.

Energy Use Intensity
67.0 kWh/yr/m²

Number PVs
88

PVs Annual Generation
118 kWh/yr/m²

PVs Contribution
8 l/s/p

EMBODIED CARBON

Embodied or embedded carbon is the carbon dioxide emitted in producing materials and products. It is a total of the energy used to extract and transport raw materials in addition to emissions associated with the manufacturing process. In construction embodied carbon is measured over the whole lifespan of a building, typically assumed to be 60 years. The carbon impact of the building's construction is calculated at each phase of the development chronologically. A1-A5 Upfront, production and construction, B1-B6 in use, and C1-C4 End of life. The carbon emissions associated with each stage are reported as kgCO2e/m². Biogenic carbon is the carbon stored within the timber elements. recycling or disposal as waste. The carbon emissions associated with each stage are reported as kgCO2e/m².

Substructure
73

Structure
96

Envelope
80

Internal fit out
64

MEP
132

A1-A5 Carbon

A1-A5

445

kgC02e/m²

UPFRONT EMBODIED CARBON

A1-A5 of 445 kgCO2e/m² excluding sequestration and 295 kgCO2e/m² including sequestration is less than the required DfE target of 525kgCo2e/m2 and 325 kgCo2e/m2 including sequestration

A-C

897

kgC02e/m²

WHOLE LIFE EMBODIED CARBON

A-C including sequestered of 897 kgCO2e/m²

This is higher than the 2025 RIBA A-C target for completed buildings of <675 kgCO2e/m². Due to the high level of services in this building.

-167

kgC02e/m²

BIOGENIC SEQUESTERED CARBON

There is 58 TCO2e biogenic carbon sequestered or 167 kgCO2e/m²

This is equivalent to 38% of the upfront embodied carbon.

DEMAND

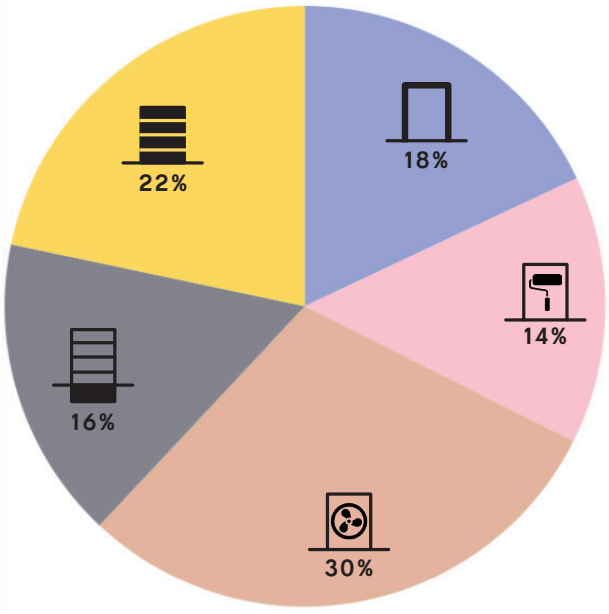
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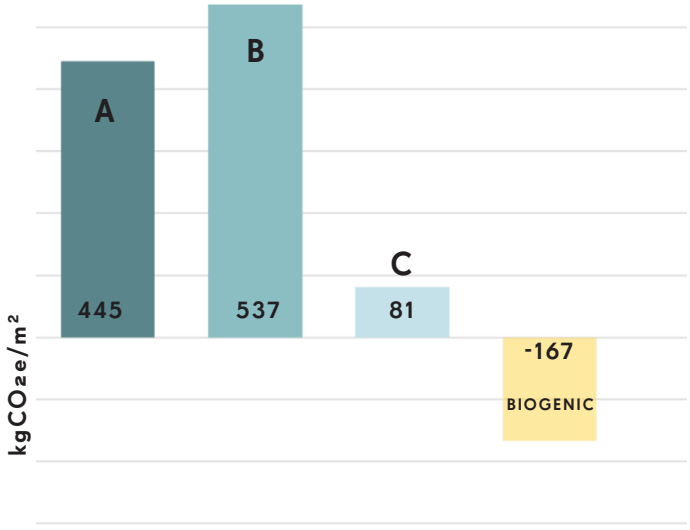
kWh/m²/y

OPERATIONAL ENERGY

The annual operational energy demand for the STEM building is 67 kWh/m²/y.

This is slightly higher than the 2030 RIBA target for educational buildings of <60 kWh/m²/y and the DfE Technical Annex J Target of 61 kwh/yr/m², this is due high level of services in this building which is not comparable to a typical school where this function would be a smaller proportion of the school.





TOTAL = 897 kgco2e/m²

Figure: Breakdown of upfront carbon A1-A5 per building elements & Breakdown of whole life carbon A-C